

Paper Manager pack — Distilling Motion Planner / MoPA-PD (Dec 9 · Alternative approaches)

Paper: Distilling Motion Planner Augmented Policies into Visual Control Policies for Robot Manipulation

arXiv: <https://arxiv.org/abs/2111.06383> · [project/code](#) · arXiv **2111.06383v1** (11 Nov 2021) · **CoRL 2021**

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PDF: [/workspace/papers/distilling-motion-planner-2111.06383.pdf](#) (= [/workspace/distill-mopa-quiet/paper.pdf](#)) · **16 pp** · ~10.5 MB

Session: Dec 9 · due 9:30 AM · Type: **Alternative approaches**

Role weights: **ALL EMPTY** — **still all five produced at balanced depth** (~equal Archaeologist / Stakeholder / Scientific Reviewer / Empiricist / Visionary); **Club Pack** (no LONGEST seat; no weight-3 over-index)

Kanta role: Produce **balanced** five-role Club Pack; deepen privileged-MP → vision-student spine + Empiricist Ablation Bot smokes; Visionary → **BEHAVIOR obstructed household; skip**

Discord/Drive

Themes: privileged classical **RRT-Connect** at train · **MoPA-RL** teacher · visual **BC smoothing** · asymmetric SAC · **planner-free** pixel policy · shorter paths than expert · **DR zero-shot** distractors (**sim-to-sim**) · **CLEAR parent** Yamada MoPA-RL 2020 · **NOT** ancestor of ACT/DP/VLA/JEPA

Sources: Summarizer ([/workspace/papers/distilling-motion-planner-2111.06383-summary.md](#)) + Senior Research ([/workspace/distill-mopa-quiet/DELIVERABLE.md](#)) + Ablation Bot ([/workspace/papers/distilling-motion-planner-2111.06383-ablations.md](#) — **AUTHORITATIVE**) + [paper_notes.md](#) / [env_outline/](#)

Sibling links: [/workspace/dart-quiet/](#) · [/workspace/diffusion-policy-quiet/](#) · [/workspace/act-aloha-quiet/](#) · [/workspace/robogen-quiet/](#) · [/workspace/leworldmodel-quiet/](#) · [/workspace/behavior-1st-place-summary.md](#)

1. One-liner

Privileged classical MP (RRT-Connect) + state RL at train → distill via visual BC smoothing + BC-guided asymmetric vision SAC into a planner-free pixel policy that beats the privileged teacher on path length, hits ~100% ASR on three obstructed Sawyer tasks, and **DR zero-shots** distractors (**sim-to-sim** only).

Contrast poles (consistency triangle — CLEAR labels; do not invent ancestry):

Pole	What they do	Relation to MoPA-PD
Classical MP (RRT-Connect / OMPL)	Collision-free joint paths	CLEAR: train-time inside MoPA-RL expert; removed at MoPA-PD deploy
MoPA-RL (Yamada et al. CoRL 2020)	State SAC + MP-augmented action space	CLEAR parent being distilled
Visual BC / CoL LfD	Imitate demos / IL+RL mix	Stage-1 baseline / foil that fails or stays long-AEL

Pole	What they do	Relation to MoPA-PD
ACT / Diffusion Policy	Chunked / diffusion visuomotor from teleop demos	CLEAR contrast — different data source & architecture; not MoPA-PD children
VLA / FM (π_0, OpenVLA, OXE)	Large multi-task VLA from heterogeneous demos	CLEAR contrast — NOT ancestor; no cited chain
JEPA / LeWM / Cosmos	Latent / generative world models	CLEAR contrast poles only — do not place MoPA-PD in JEPA/Cosmos genealogy
DART / DAgger	Covariate-shift recovery (noise / query)	Shared vocabulary (Ross et al.); MoPA-PD fix = BC $\rightarrow$ online RL + expert buffer
RoboGen / RelMoGen	MP primitives online in generative sim	Opposite deploy story — they keep MP; MoPA-PD amortizes it away

Do NOT invent / claim: MoPA-PD as ancestor of ACT/DP/ π_0 /OpenVLA/JEPA/Cosmos; sim-to-real success (paper = **sim-to-sim** DR); course smoke = Tab.1 5-seed Assembly **100%**; “vision-only learning” erasing privileged **asymmetric critic** at train; Discord/Drive checklist.

2. Summary (exec overview + key numbers)

Liu* & Uppal* et al. (CoRL 2021 / arXiv 2111.06383) attack obstructed-scene visuomotor learning: sparse exploration around obstacles + high-dim vision. Prior **MoPA-RL** (Yamada et al. CoRL 2020) solves exploration with a **privileged geometric-state** planner (RRT-Connect for large Δq ; direct SAC for contact-rich small moves), but that state and planner compute are unavailable at real deploy. **MoPA-PD** throws the planner and full geometric state away at inference via two stages:

1. **Visual BC** on MoPA-RL low-level (direct-A) trajectories — removes MP dependency **and** smooths jittery RRT paths.
2. **Vision RL** (asymmetric actor-critic SAC) guided by **smoothed BC** trajectories in expert buffer Re; actor $\leftarrow$ BC, critic $\leftarrow$ MoPA-RL Q; **1:3** Re:R π mix; **small α** .

Why Alternative approaches / Dec 9: Teaches a durable **hybrid classical $\leftrightarrow$ learned** pattern — use MP when geometry is known in sim; ship a visual policy that no longer needs it. Antecedent / alternative to “VLA-from-demo only,” **not** a child of foundation-model stacks.

Axis	Number / claim
Venue / code	CoRL 2021 ; arXiv 2111.06383v1 ; https://clvrai.com/mopa-pd CLEAR
Expert parent	MoPA-RL Yamada et al. CoRL 2020 (CLEAR)
Planner	RRT-Connect (Kuffner & LaValle) — train-only inside MoPA-RL
Obs / robot	MuJoCo Sawyer 7-DoF ; RGB 32$\times$32 ; horizon 250 ; γ 0.99
Budget	MoPA-RL 1M + visual 2M = 3M env steps; 5 seeds $\times$ 100 eps
Ours Push (Tab.1 ASR/AEL/ADR)	100.0 / 32.0 / 110.8
Ours Lift	99.0 / 42.0 / 101.7
Ours Assembly	100.0 / 61.7 / 84.5
Expert MoPA-RL Push AEL	111.0 $\rightarrow$ Ours 32.0 ($\sim 3.5\times$ shorter)
BC-Visual Lift ASR	62.0 (shows Stage-2 necessity)

Axis	Number / claim
w/o BC smoothing Assembly	ASR 0.0 (with: 100.0)
MoPA-Asym. SAC / Asym. SAC	0.0 ASR all (or near-zero)
DR zero-shot (Tab.2)	ASR >96% all; worst Lift Scen.1 96.7
Sub-opt teacher (Tab.4)	Lift @0.5M MoPA 41.4 → Ours 99.0 ; Push @0.2M 69.2 → 100
BC wall-clock (Tab.8)	Push 30 / Lift 39 / Asm 120 min
Scope	Sim-only ; real-robot / vision-critic FT = future work

Must-cite tattoo: Push **100 / 32.0 / 110.8** · Lift **99 / 42 / 101.7** · Assembly **100 / 61.7 / 84.5** · expert Push AEL **111** → **32** · Assembly ASR **0** w/o BC smooth · DR **≥96.7** · **refuse sim-to-real laundering.**

Do NOT claim: course smoke = paper Tab.1 5-seed Assembly 100%; DR distractors = sim-to-real; MoPA-PD ancestry of modern VLA/JEPA; “vision-only” without noting privileged critic; full 3M×5×3 from Push-lite Partial.

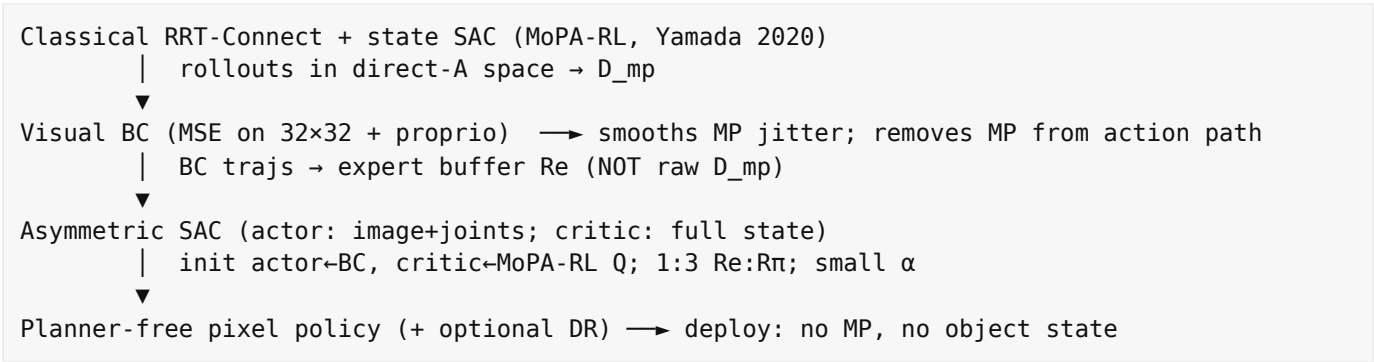
3. Keywords

MoPA-PD; Motion Planner Augmented Policy Distillation; MoPA-RL; RRT-Connect; visual behavioral cloning; trajectory smoothing; asymmetric actor-critic; SAC; expert replay buffer; entropy α ; domain randomization; obstructed manipulation; Sawyer Push / Lift / Assembly; privileged train-time planning; planner-free deploy; Alternative approaches; BEHAVIOR clutter transit (TENTATIVE recipe)

4. Deep dive — Privileged MP → planner-free vision (PRIMARY SESSION CONTRIBUTION)

Callout: Scientific product of *this* PDF = a **distillation recipe** that amortizes classical motion planning into a reactive visual controller. Session type = **Alternative approaches** because Dec 9 must teach **why classical robotics remains a training superpower** even when the shipped policy must be vision-only — **without** claiming MoPA-PD founded ACT/DP/VLA.

Pipeline (Fig. 1 moral)



Components privilege table

Component	Role	Privilege / deploy
RRT-Connect	Collision-free large Δq inside MoPA-RL	Train-only (via expert)
MoPA-RL SAC (state)	Expert exploring obstructed scenes	Train-only; D_mp + critic weights
Visual BC	Distill + smooth ; actor init	Train; produces Re
Expert buffer Re	Guided exploration / anti-forgetting	Train; BC traj , not raw MP
Agent buffer R_π	SAC experience	Train
Asymmetric critic $Q_\psi(s,a)$	Value under full state	Train-only privilege
Visual actor $\pi_\theta(o, \text{joints})$	Deployed control	Deploy (no MP, no object state)
Domain randomization	Appearance invariance	Train; enables zero-shot distractors
Entropy α (small)	Exploit privileged prior	Train hyperparam

What is “alternative” here — CLEAR

Claim	Status
MoPA-PD distills MoPA-RL (Yamada CoRL 2020)	CLEAR
Train-time RRT-Connect ; removed at deploy	CLEAR
BC smoothing necessary (Assembly ablation ASR 0 $\rightarrow$ 100)	CLEAR
Asymmetric actor-critic (Pinto et al.) used	CLEAR
Beats teacher on AEL (Push 111 $\rightarrow$ 32)	CLEAR
DR zero-shot distractors >96% ASR (sim-to-sim)	CLEAR
Contrast to ACT / DP / VLAs as different poles	CLEAR (session genealogy)
MoPA-PD is ancestor of π_0 / OpenVLA / JEPA / Cosmos	FALSE — do not claim
Real-robot MoPA-PD success	Not in paper (future work)
Official BEHAVIOR’26 baseline	TENTATIVE / unverified

Architecture checklist (Empiricist pin — App. B)

Piece	Spec (paper)
MoPA-RL nets	3× FC 256 , ReLU
Visual actor / BC	3-layer CNN $\rightarrow$ 3× FC 256 LeakyReLU $\rightarrow$ Gaussian head
SAC	Adam lr 1e-5 , γ 0.99 , buffer 10⁶ , batch 256 , image 32×32
BC	Adam lr 5e-4 , $\sim$ 1M pairs, 9:1 split, batch 512 ; pick earliest epoch with 100% val success
Sample mix	Re : R_π = 1:3
Reward scales	Push/Lift/Asm 0.8 / 0.5 / 1.0 ; success thresh 0.05 ; bonus +150
Action dims	Push 7 / Lift 8 / Asm 7

Exam bite: "MoPA-PD = MoPA-RL expert (RRT-Connect+SAC) $\rightarrow$ visual BC smooths jitter $\rightarrow$ asym. SAC with BC buffer + weight init + small α $\rightarrow$ $\sim$ 100% Sawyer clutter, shorter than expert (Push AEL **111** $\rightarrow$ **32**), DR zero-shot distractors (**sim-to-sim**); MP gone at deploy; **CLEAR parent** Yamada 2020; **NOT** ACT/DP/VLA/JEPA ancestor."

5. Deep dive — Ablations (Ablation Bot AUTHORITATIVE)

Callout: Ablation Bot extract **AUTHORITATIVE** at /workspace/papers/distilling-motion-planner-2111.06383-ablations.md. Focus ranks: (1) planner-on vs distill, (2) BC vs MoPA-PD, (3) obstructed SR, (4) domain randomization. Course Empiricist: **Y #1/#2/#4; Partial #3/#5**; cite **#6/#7**; **N** full 3-task×5-seed×3M.

Rank	Factor	Paper numbers	Course smoke
1	Planner-on vs distill (Tab.1 / Fig.3)	Ours ASR 100/99/100 ; MoPA Asym 0/0/0 ; MoPA-RL 98.2/95.0/99.8 ; Push AEL 32 vs 111	Y — RC1 PRIMARY
2	BC-Visual vs Ours	BC ASR 99.4/62.0/97.0 vs 100/99/100 ; BC AEL ~108–118 vs 32/42/61.7	Y — RC2 PRIMARY (1 task)
3	Domain-rand zero-shot (Tab.2)	ASR >96% ; Orig 99.7/99.3/100 ; S1 100/96.7/100 ; S2 99.3/97.3/100	Partial — RC4
4	BC trajectory smoothing on/off	w/o Asm 0.0 ; with 100.0	Y — RC3 PRIMARY
5	Actor-critic weight init	w/o: ~0 SR till 1.2M ; with: ≈1.0 before 1.0M	Partial — RC5
6	Entropy α	smaller $\log(\alpha)$ faster/stabler; teacher $\log(\alpha)$ Push –0.63 / Lift –2.7 / Asm –0.29	cite / optional
7	Sub-optimal MoPA expert (Tab.4)	Lift 41.4 → 99.0 ; Push 69.2 → 100	Partial / cite
Defer	Full 3M×5×3 + long MoPA-RL retrain; real-robot	—	N

Min Empiricist package: RC0 → **RC1 (#1) + RC2 (#2) + RC3 (#4)** → Partial RC4/RC5 → cite #6/#7. Log **ASR, AEL, ADR, EE jerk, MP_calls_at_infer=0, Re source, init, log(α), peak VRAM**.

Recipe exam bite: "Smoothing is load-bearing (Asm **0** without it); MoPA-Asym SAC = **0** proves 'just put MP in the visual loop' fails; distillation collapses MP into a reactive policy — that *is* the point."

6. Role-ready sections (all five — **BALANCED Club Pack**)

Balanced pack. Equal depth across five roles. No seat weight-3. Ablation Bot focus areas (planner-on vs distill, BC vs MoPA-PD, obstructed SR, domain-rand) shared across Empiricist RCs + Reviewer/Archaeologist honesty.

6.1 Archaeologist — balanced (~12–14)

Seat charge: Place MoPA-PD carefully in classical-MP ↔ learned-control history. **CLEAR parent = MoPA-RL Yamada 2020**. Fence false FM/JEPA ancestry. Equal depth with other four.

Priors — CLEAR

Prior	Role	Label
MoPA-RL Yamada et al. CoRL 2020	Direct parent expert	CLEAR

Prior	Role	Label
RRT-Connect Kuffner & LaValle 2000	Collision-free large moves	CLEAR
Asymmetric actor-critic Pinto et al. RSS 2018	Vision actor / state critic	CLEAR
SAC Haarnoja et al. ICML 2018	RL backbone	CLEAR
BC / ALVINN Pomerleau 1989; covariate shift Ross et al. 2011	Stage-1 + diagnosis	CLEAR
RelMoGen / PBCS; neural MP (Qureshi, Strudel, Jurgenson & Tamar)	Related MP+RL / MP distill	CLEAR related
CoL (Goecks et al. 2020); DQfD; DAPG	LfD baselines that fail here	CLEAR
IKEA furniture env Lee et al.; Tobin DR	Task / transfer priors	CLEAR

Genealogy (CLEAR framing — no false FM lineage)

```

Classical MP (RRT-Connect)
+ state SAC in clutter (MoPA-RL, Yamada 2020)
  → privileged expert demos + critic
    → visual BC (smooth + remove MP)
      → BC-guided asymmetric vision SAC (MoPA-PD)
        → planner-free pixel policy (+ DR zero-shot sim-to-sim)

```

Subsequent / contrast — label carefully

Item	Note	Label
ACT / Diffusion Policy	Teleop-demo visuomotor IL; different pole	CLEAR contrast
π_0 / OpenVLA / OXE	Large VLA; no cited ancestry from MoPA-PD	FALSE ancestry — do not claim
LeWM / JEPA / Cosmos	Latent / generative WM poles	CLEAR contrast only
RoboGen MP menu	Keeps MP online — opposite deploy	CLEAR parallel / contrast
Soft influence on “privileged sim → vision deploy” recipes	Pattern family; verify per paper	TENTATIVE

Archaeologist one-liner: *Classical MP is the privileged teacher, not the product — MoPA-PD is the CoRL’21 bridge from Yamada MoPA-RL to planner-free vision, and a contrast ancestor/alternative to teleop-demo VLAs, not a JEPA/Cosmos child.*

Archaeologist exam bite: "MoPA-PD = RRT-Connect MoPA-RL → BC smooth → asym SAC → planner-free vision; **CLEAR parent** Yamada 2020; Push AEL **111** → **32**; Asm **0** w/o smooth; **NOT** ACT/DP/VLA/JEPA ancestor."

6.2 Stakeholder — balanced (~12–14)

Seat charge: Why schedule in Alternative approaches; buy the **pattern**, not the 2021 Sawyer checkpoint.

Buy / why schedule Dec 9:

- Locks **privileged-MP** → **vision student** before pure VLA weeks.

- Clean answer to “why not only VLA-from-demo?”: in **heavily obstructed sim**, privileged MP exploration is what makes demos exist; distillation is the productization step.
- HIGH motion-planning / obstruction hook without a full TAMP stack.
- Hard numbers: Push **100/32/110.8**, Asm **0** → **100** with smoothing, DR ≥ 96.7 .

Buy / build vs modern stacks

Situation	Recommendation
Sim has accurate geometry; need vision deploy; clutter is the hard part	Buy the MoPA-PD pattern: MP/TAMP or MoPA-style teacher → BC smooth → vision RL/IL
Abundant real teleop / OXE-scale demos; open-world language	Prefer ACT / DP / VLA ; MoPA-PD complementary sim recipe
No reliable collision geometry even in sim	Stage-0 dies — don’t force it
Need foundation-model generalization across embodiments	MoPA-PD is narrow Sawyer MuJoCo — use as idea , not checkpoint

Risks / costs:

- Sim-only; authors flag real fine-tune / vision-critic as future work — **sim-to-real tax unpaid**.
- 32×32 images — not modern wrist/cam stacks.
- Depends on working MoPA-RL (or other MP-RL) teacher; Stage-0 engineering cost.
- Asymmetric critic assumes sim state — real online RL needs a different critic story.
- Don’t pitch as competing with π_0 -scale generalists; pitch as **clutter exploration prior**.

Decision ask: Required Alternative-approaches classic for privileged-teacher distillation; pair with DART (recovery demos) and ACT/DP (visuomotor IL) as **contrast**, not ancestors. Buy the **pattern**.

Stakeholder exam bite: "Schedule MoPA-PD to lock **privileged-MP** → **vision student** before VLA weeks; buy the pattern, not the Sawyer checkpoint; police sim-to-real laundering of Tab.2."

6.3 Scientific Reviewer — balanced (~12–14)

Claims under review

1. Distilling MP-augmented state RL into vision via BC+guided RL removes MP/state at deploy.
2. BC smoothing is necessary (esp. Assembly).
3. Weight init + expert buffer + small α yield high sample efficiency vs Asym. SAC / CoL / MoPA-Asym. SAC.
4. DR enables zero-shot distractor / appearance transfer at >96% ASR (**sim-to-sim**).
5. Method can exceed expert path quality (AEL/ADR).

What holds

- Table 1 is decisive: vision-from-scratch and “MP inside visual MoPA” both **zero out**; MoPA-PD hits ~100% with **better AEL than the state expert**.
- BC-smoothing ablation is a clean causal knife (Assembly **0 vs 100**).
- Init ablation (Fig. 7) strongly supports transferring MoPA-RL critic + BC actor.
- Sub-optimal expert Table 4 shows distillation is not mere cloning.
- Related-work honesty: distinguishes neural MP papers that stay state-based / non-manip.

What is soft / threatened

- **No real-robot results** — transfer claims are **sim-to-sim** DR, not sim-to-real.
- Single embodiment (Sawyer), three tasks, **32×32** RGB — external validity to household cams unknown.
- CoL / Asym. SAC hyperparameters may be disadvantaged in clutter; still, zeros across seeds are stark.
- Asymmetric critic = lingering privilege at train; paper's own future work admits need for vision critic.
- "Outperforms SOTA" scoped to 2021 obstructed Sawyer suite — not modern visuomotor IL.

Verdict: Strong **CoRL systems/methods** paper for the privileged-teacher distillation claim in obstructed sim. Accept the mechanism + ablations. **Do not** upgrade to "solved real clutter manipulation" or "ancestor of modern VLAs."

Reviewer exam bite: Mechanism and ablations are crisp; sim-to-sim DR is real; real-world and modern-cam generalization remain open — grade as **Alternative approaches classic**, not as foundation-model prehistory. Demand **MP@infer=0** evidence, not only ASR.

6.4 Empiricist — balanced (~12-14; Ablation Bot AUTHORITATIVE)

Seat empty this session but produce full course smoke cards for Paper Manager continuity. **Balanced** — equal depth with other roles; richest logging on RC1/RC2/RC3. Ablation Bot ranks **1-7** govern order.

Claim under measurement

MoPA-PD (planner-off) beats Asym SAC / matches or beats MoPA-RL on Push ASR @ matched visual budget; AEL directionally $\ll$ teacher/BC; **MP_calls@infer = 0**; BC-smoothed $Re \geq$ raw; DR Partial holds OOD within tolerance — **directional, not** Tab.1 5-seed Assembly 100%.

Hypotheses (falsifiable; Ablation Bot ranks)

1. **H-planner (#1):** MoPA-PD $\gg$ MoPA Asym / Asym SAC on Push ASR; AEL(MoPA-PD) $\ll$ AEL(MoPA-RL demos); **MP_calls@infer = 0**.
2. **H-bc (#2):** On 1 task, MoPA-PD $>$ BC-Visual on Lift-style hardness **or** on AEL $\downarrow$ /ADR $\uparrow$ even when BC ASR is already high.
3. **H-smooth (#4):** $Re =$ BC-smoothed $>$ $Re =$ raw MoPA-RL on ASR and/or convergence; Assembly moral: raw $Re \rightarrow$ **ASR 0**.
4. **H-dr (#3 Partial):** DR-trained policy retains high ASR on original + distractor stub (Tab.2 $>96\%$ moral — directional).
5. **H-init (#5 Partial):** BC+MoPA-critic init reaches nonzero ASR far earlier than random (Fig.7).
6. **H-alpha (#6):** Smaller $\log(\alpha) \rightarrow$ faster/stabler visual SAC when teacher prior exists.
7. **H-teacher (#7):** Suboptimal MoPA teacher still yields high ASR after distill (Tab.4 Lift 41.4 $\rightarrow$ 99).

Falsifiers: MoPA-Asym $\geq$ Ours; BC AEL $\leq$ Ours when both succeed; raw $Re \geq$ smooth; DR collapses on Scen.1; random init $\approx$ BC init; large α best; distill needs perfect teacher; claiming Tab.1 Assembly 100% from Push-lite.

Prioritized matrix

#	Experiment	Ablation Bot	Smoke	Pri
RC1	planner_removal_bakeoff	#1	Y — PRIMARY	P0
RC2	bc_vs_mopapd (1 task)	#2	Y — PRIMARY	P0
RC3	bc_smoothing_ablation	#4	Y — PRIMARY	P0
RC4	domain_rand_transfer	#3	Partial	P1
RC5	weight_init_ablation	#5	Partial	P1
RC6	entropy_alpha_sweep	#6	cite / optional	P2
RC7	suboptimal_teacher	#7	Partial / cite	P2
—	Full 3M×5×3; real-robot; fresh long MoPA-RL	—	N	Skip

Min package: RC0 → **RC1 + RC2 + RC3** → Partial RC4/RC5 → cite #6/#7.

Exact reproduce stack (EDIT)

```
# Official
# clone / pin https://clvr.ai.com/mopa-pd (+ Yamada MoPA-RL deps); prefer frozen demos
export MOPA_PD_WORK_ROOT=$SCRATCH/mopa-pd-smoke
cd /workspace/distill-mopa-quiet
bash env_outline/setup_mamba.sh
# mamba activate mopa-pd-scout
mkdir -p logs runs checkpoints demos
# Outlines only – do NOT sbatch from quiet box
# NEVER claim Tab.1 Assembly 100% / DR≥96.7 absolute / sim-to-real from Push-lite smoke
# ALWAYS log MP_calls_at_infer for MoPA-PD (=0)
```

Run cards (from DELIVERABLE)

RC0 — demo_ingest_smoke

Field	Spec
Goal	Load/collect MoPA-RL (or stand-in) demos → (s, o, a, r, s') + 32×32 RGB; direct-A labels
Pass	Schema OK; 0 NaNs; wall <1 h
Script	env_outline/slurm_demo_ingest_smoke.sh

RC1 — planner_removal_bakeoff (#1 — PRIMARY)

Field	Spec
Paper	Tab.1 — Ours 100/99/100 ; MoPA Asym 0/0/0 ; Push AEL 32 vs 111
Goal	Matched-budget Push: MoPA-PD vs Asym SAC (+ opt MoPA-Asym); prove planner-off distill wins
Hard req	Log MP_calls_at_infer=0 for MoPA-PD
Pass	MoPA-PD ASR >> Asym SAC; AEL directionally << teacher/BC
Script	env_outline/slurm_baseline_planner_removal.sh

RC2 — bc_vs_mopapd (#2 — PRIMARY)

Field	Spec
Paper	BC ASR 99.4/62.0/97.0 vs Ours 100/99/100 ; BC AEL ~ 108–118 vs 32/42/61.7
Goal	1 task head-to-head; prioritize AEL/ADR (and Lift if budget)
Pass	MoPA-PD AEL < BC AEL when both succeed or MoPA-PD ASR $\gg$ BC on Lift-lite
Script	env_outline/slurm_bc_vs_mopapd.sh

RC3 — bc_smoothing_ablation (#4 — PRIMARY)

Field	Spec
Paper	w/o smoothing Assembly 0 ; with 100 ; Fig.8 jitter
Goal	Re=BC vs Re=raw; Push primary; Assembly cite/stub
Pass	Smooth $\geq$ raw on ASR or speed; jerk $\downarrow$
Script	env_outline/slurm_bc_smoothing_ablation.sh

RC4 / RC5 — Partial DR / init

Card	Spec
RC4 DR	Short DR Push; eval original + distractor stub; OOD within ~10–15 pp of ID (directional) — slurm_domain_rand_transfer.sh
RC5 init	{init: on\ off} @ matched Push; init dominates early ASR — slurm_weight_init_ablation.sh

Numbers to memorize (exam)

Fact	Value
Ours Push / Lift / Asm	100/32.0/110.8 · 99/42/101.7 · 100/61.7/84.5
Expert Push AEL	111 → 32
w/o BC smooth Asm	ASR 0.0
MoPA-Asym / Asym SAC	0 ASR
BC Lift ASR	62.0
DR worst cell	Lift Scen.1 96.7
Sub-opt Lift	MoPA 41.4 → Ours 99.0
Scope	Course = Push lite; sim-to-sim DR; no sim-to-real laundering

GPU / RAM / TIME honesty (course)

Workload	VRAM	Wall (order)	Smoke
Demo ingest + BC	4–12 GB	0.5–2 h	Y
RC3 smoothing Push	8–16 GB	4–12 h	Y
RC1 planner-removal	8–16 GB	8–20 h	Y
RC2 BC vs MoPA-PD	8–16 GB	4–16 h	Y

Workload	VRAM	Wall (order)	Smoke
RC4/RC5 Partial	8–16 GB	4–16 h	Partial
Full 3M×5×3 + Assembly	16–24 GB+	≫2–5 days	N
Fresh MoPA-RL + RRT from scratch	8–16 GB + CPU MP	+1–3 days	N if demos exist
Real-robot	—	—	N

Empiricist exam bite: "Quote **100/32/110.8, Assembly 0 without BC smooth**, $DR \geq 96.7$; RC1+RC2+RC3 = planner-off + BC bake-off + smoothing with **MP@infer=0**; refuse to launder sim-to-sim distractors into sim-to-real; Ablation Bot AUTHORITATIVE ranks 1–7."

6.5 Visionary — balanced (~12–14) — BEHAVIOR obstructed household

Seat charge: Decision-oriented **BEHAVIOR obstructed household** hooks. Privileged-info-at-train → reactive vision deploy. Pair with DART recovery. **No Discord/Drive checklist.**

BEHAVIOR connection (honest)

BEHAVIOR-class household activities are dense with **obstruction, clutter, and multi-stage contact**. MoPA-PD's durable idea is not the Sawyer checkpoint — it is:

When sim geometry is known, use classical MP (or TAMP) as a privileged explorer/teacher; distill to a vision policy that runs without that privilege.

BEHAVIOR need	MoPA-PD-shaped response
Collision-heavy rooms	Train-time RRT/BIT*/TAMP teachers for reach/place primitives
Vision-only deploy	Asymmetric train / vision actor deploy; then real fine-tune
Long / multi-stage activities	Stage-wise teachers + BC smooth before vision RL/VLA distill
Distractors / appearance shift	Keep DR (or modern DA) — but Tab.2 ≠ layout/clutter OOD
Hybrid with modern policies	MP-teacher demos as one column beside teleop/VLA demos — don't replace $\pi_{0.5}$ with 32×32 SAC

Visionary bets (balanced; not monopoly)

- 1. Sim-privileged planner teacher → deploy-reactive VLA:** In OmniGibson / BEHAVIOR-sim, run privileged MP/MoPA-like teacher for hard clutter transit; distill into $\pi_{0.5}$ / N1.7 / DP with **BC then light RL/FT**.
- 2. Mandatory smoothing card:** Never dump raw RRT waypoints into Re; BC or trajectory-optimize first; log jerk.
- 3. Privileged critic / asymmetric train only:** Keep state critic in sim FT; **forbid** state at BEHAVIOR eval — reviewer's MP@infer=0 analog.
- 4. Pair with DART recovery:** Planner-distill covers **transit**; DART-style recovery teleop covers **slip/miss**.
- 5. Do not ship online RRT on the robot** for the leaderboard unless latency allows; MoPA-PD's point is **amortize**.

6. **DR is not enough:** Appearance DR $\neq$ layout/clutter OOD — add instance / furniture variation.
7. **Metric honesty:** Report collision rate, path length, stage success — not only final ASR.
8. **Substrate:** Race on $\pi_{0.5}$ / GR00T N1.7; treat MoPA-PD as **distill recipe**, not a 32×32 reimplementation.
9. **Empiricist honesty:** Push-lite validates smoothing/init/planner-removal; transfer claims need BEHAVIOR-sim Partial — never Tab.1 Assembly 100% as proxy.
10. **One narrative line:** MoPA taught **plan when the action is big**; MoPA-PD taught **practice the plan until the reactive visual policy no longer needs the planner**; 2026 should **distill privileged clutter transit into VLA deploy policies**, then **rehearse recoveries** (DART) where distill still fails.

Follow-up research

1. Replace 32×32 CNN with modern encoders; keep privileged MP teacher.
2. Vision critic / self-supervised critic for real asymmetric $\rightarrow$ symmetric transition.
3. MP-teacher demos into RLDS/OXE-schema shards labeled `teacher=mopa|rrt`.
4. Combine BC-smooth with action chunking (ACT/DP) on planner-taught traj.
5. Measure whether privileged-MP demos help BEHAVIOR obstruction categories more than equal hours of teleop.
6. Planner jitter metrics as first-class data-quality filters before IL.

New apps (beyond Discord/Drive)

- Furniture / cabinet insert curricula in household sim.
- Cluttered fridge / shelf reach-behind with MP teachers.
- Synthetic hard-negative obstacle layouts for VLA eval.
- Latency-critical deploy: bake away RRT online cost into a reactive visual policy.

(No Discord / Drive checklist content.)

Visionary one-liner: *For BEHAVIOR obstructed households, keep classical MP as a sim teacher, distill (BC-smooth + modern IL/RL/VLA heads) into vision students — MoPA-PD is the 2021 proof that the teacher can disappear at deploy without disappearing from the curriculum.*

Visionary exam bite: "MoPA-PD = **planner-distill recipe** for clutter transit beside **recovery-demo** (DART) and modern VLA substrates; cite Push **100/32/110.8**, Asm **0** w/o smooth, DR $\geq$ **96.7** as morals; Empiricist smokes directional only; **no Discord/Drive**."

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export MOPA_PD_WORK_ROOT=$SCRATCH/mopa-pd-smoke
cd /workspace/distill-mopa-quiet
bash env_outline/setup_mamba.sh
# mamba activate mopa-pd-scout
mkdir -p logs runs checkpoints demos
# EDIT: pin clvrai.com/mopa-pd + MoPA-RL; prefer frozen demos
# Outlines only — do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	mamba sketch (Torch + MuJoCo + optional MoPA/RRT)
env_outline/setup_mamba.sh	env create; MuJoCo EGL/OSMesa notes
env_outline/run_checklist.md	GPU / TIME honesty preflight
env_outline/README.md	cluster usage
env_outline/slurm_demo_ingest_smoke.sh	RC0
env_outline/slurm_baseline_planner_removal.sh	RC1 / #1 PRIMARY
env_outline/slurm_bc_vs_mopapd.sh	RC2 / #2 PRIMARY
env_outline/slurm_bc_smoothing_ablation.sh	RC3 / #4 PRIMARY
env_outline/slurm_domain_rand_transfer.sh	RC4 / #3 Partial
env_outline/slurm_weight_init_ablation.sh	RC5 / #5 Partial
env_outline/slurm_entropy_alpha_sweep.sh	RC6 / #6 optional
env_outline/slurm_suboptimal_teacher.sh	RC7 / #7 Partial

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
Distill removes MP + state at infer	MoPA-PD eval with MP_calls=0 and image+joint obs only	Bit-identical RRT teacher
BC smoothing fixes jitter / hard tasks	RC3: smooth Re $\geq$ raw on Push ASR/speed; jerk $\downarrow$	Assembly 100% from smoke
Weight init for sample-efficiency	RC5: init $\gg$ random early ASR	Exact Fig.7 1.2M cutoff
Low α helps with teacher prior	Cite / optional RC6	Full Fig.5 grid
Beats pure visual RL / visual+MP / naive LfD	RC1: MoPA-PD $\gg$ Asym SAC; AEL < BC	CoL full reimpl; MoPA-Asym if MP broken
Beats privileged MoPA on AEL/ADR	Directional AEL $\downarrow$ vs BC and teacher demos	Tab.1 absolute ADR 110.8
Suboptimal teacher distillable	RC7 Partial directional	Tab.4 full
DR zero-shot $\geq 96\%$ ASR	RC4 Partial: OOD holds within tolerance	Absolute $\geq 96\%$ / Scen.2 furniture
Full paper protocol	Document N	Never course

Pass: documented smoke with Ablation Bot **#1 + #2 + #4** confirmed directionally + **MP@infer=0** + logged knobs.

Fail / overclaim: “reproduced CoRL Tab.1 5-seed Assembly 100%” / “sim-to-real” from Push lite Partial.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Archaeologist	balanced · ~12-14	CLEAR/TENTATIVE; MoPA-RL parent; false FM ancestry fence; genealogy; exam bite
Stakeholder	balanced · ~12-14	Buy pattern not checkpoint; risks (sim-only, 32×32); decision ask
Scientific Reviewer	balanced · ~12-14	Tab.1+smoothing+init; verdict Accept Alternative classic; fence sim-to-real / VLA ancestry
Empiricist	balanced · ~12-14	H-planner/bc/smooth; RC1+RC2+RC3; GPU honesty; must-cite tattoo; exam bite
Visionary	balanced · ~12-14	BEHAVIOR obstructed household; planner-distill + DART; no Discord/Drive

Spine (paste into Slide Maker) — 5 bullets:

1. **Privileged classical MP (RRT-Connect) at train** → **visual BC smooth** → **asym SAC** → **planner-free pixel policy**; shorter paths than expert; DR zero-shot distractors (**sim-to-sim**).
2. **Must-cite**: Push **100/32.0/110.8**; Lift **99/42/101.7**; Assembly **100/61.7/84.5**; expert Push AEL **111** → **32**; Assembly ASR **0** w/o BC smooth; DR **≥96.7**; **refuse sim-to-real laundering**.
3. **Taxonomy**: classical MP | **MoPA-RL parent (Yamada 2020)** | MoPA-PD distill | ACT/DP/VLA (**CLEAR contrast, not children**) | JEPA/LeWM/Cosmos (**contrast poles only**).
4. **Lineage (Club Pack — all seats empty, balanced)**: CLEAR parent MoPA-RL; inventum = BC-smooth + asym SAC distill that **removes** MP at deploy and **beats** teacher AEL; **NOT** ancestor of ACT/DP/VLA/JEPA.
5. **BEHAVIOR'26 + honesty**: MoPA-PD = **planner-distill recipe** for clutter transit beside DART recovery + modern VLA substrates; Empiricist RC1-RC3 directional; **do not invent** sim-to-real / Assembly 100% from Push smoke — **no Discord/Drive**.

GitHub title sketch: [ECE 605] - Distilling Motion Planner / MoPA-PD Making Club Pack

Full Slide Maker brief: /workspace/distill-mopa-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/distill-mopa-quiet/KANTA_PACK.md
PDF	/workspace/distill-mopa-quiet/KANTA_PACK.pdf
Slide brief	/workspace/distill-mopa-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/distill-mopa-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/distilling-motion-planner-2111.06383-summary.md
Ablations	AUTHORITATIVE — /workspace/papers/distilling-motion-planner-2111.06383-ablations.md
Paper notes	/workspace/distill-mopa-quiet/paper_notes.md

Artifact	Path
Env outlines	/workspace/distill-mopa-quiet/env_outline/
Paper PDF	/workspace/papers/distilling-motion-planner-2111.06383.pdf
Text extract	/workspace/papers/distilling-motion-planner-2111.06383.txt · paper_clean.txt
Page figs	/workspace/papers/mopa-pd-figs/page-01.png ... page-16.png
Sibling links	/workspace/dart-quiet/ · /workspace/diffusion-policy-quiet/ · /workspace/act-aloha-quiet/ · /workspace/robogen-quiet/ · /workspace/leworldmodel-quiet/
Code	https://clvrai.com/mopa-pd

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 — Dec 9 Alternative approaches · ALL seats EMPTY · balanced five-role Club Pack · Ablation Bot AUTHORITATIVE ranks 1–7 · CLEAR parent MoPA-RL Yamada 2020 · NOT ACT/DP/VLA/JEPA ancestor · must-cite Push 100/32.0/110.8 · Lift 99/42/101.7 · Assembly 100/61.7/84.5 · expert Push AEL 111 → 32 · Assembly ASR 0 w/o BC smooth · DR ≥96.7 · refuse sim-to-real laundering · Visionary BEHAVIOR obstructed household · no Discord/Drive · outlines only — no cluster execution.